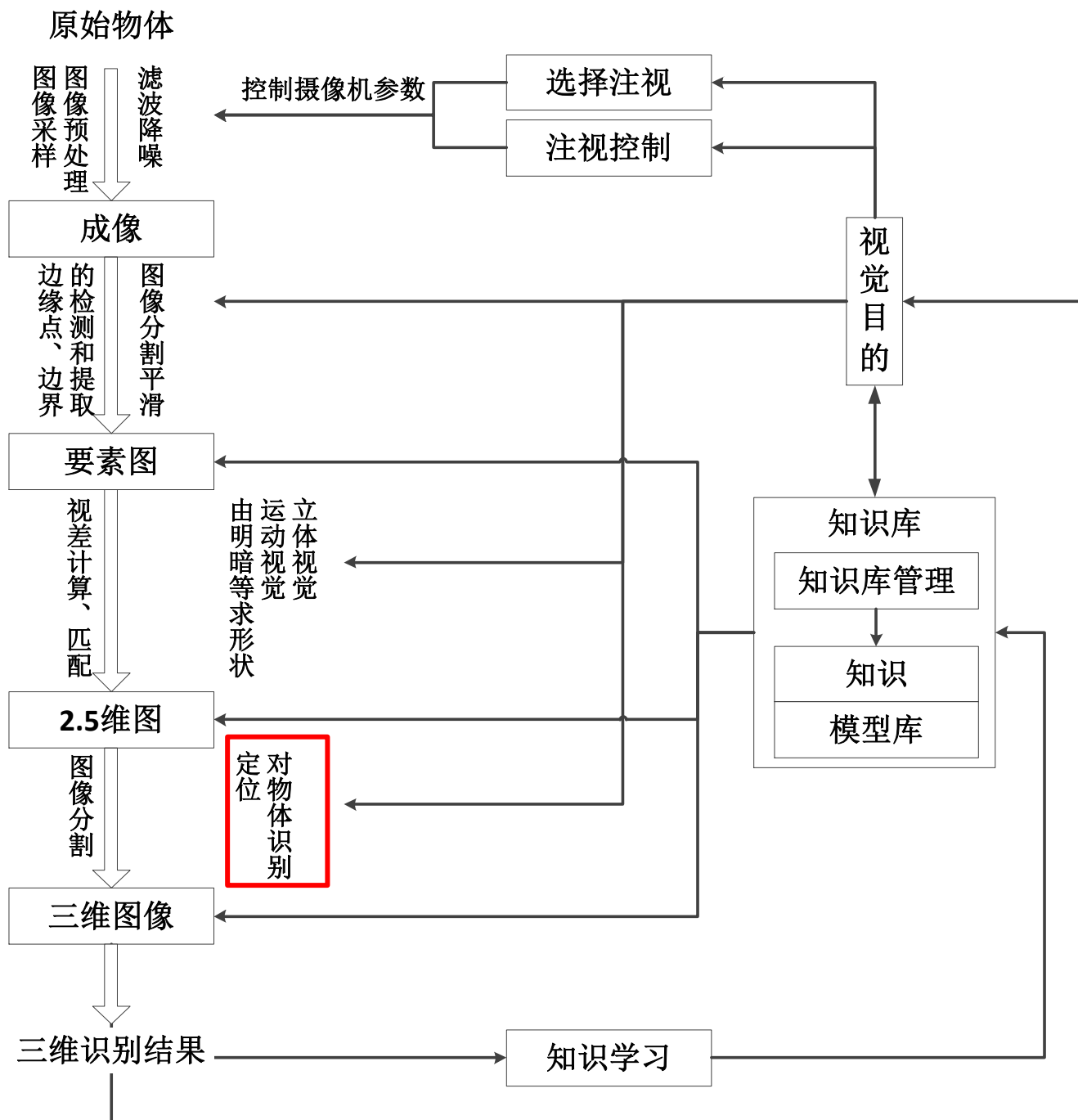
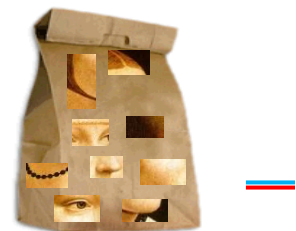


计算机视觉 ——目标识别

2026年春季

桑农，王岳环





目的

- 了解什么是目标识别？
- 熟悉两种目标识别方法
- 可在编程作业中应用



目标识别

- 目标检测
 - 人脸检测（Haar+AdaBoost）
 - 行人检测
- 目标分类
 - 局部特征（bag-of-words, 空间金字塔）
- 目标身份确认
 - 人脸 身份确认
 - 行人身份确认



目标识别

■ 什么是目标识别？

◆ What is it?

Object classification



◆ Where is it?

Object detection

◆ Who is it?

◆ Object Identification



目标识别包括什么？



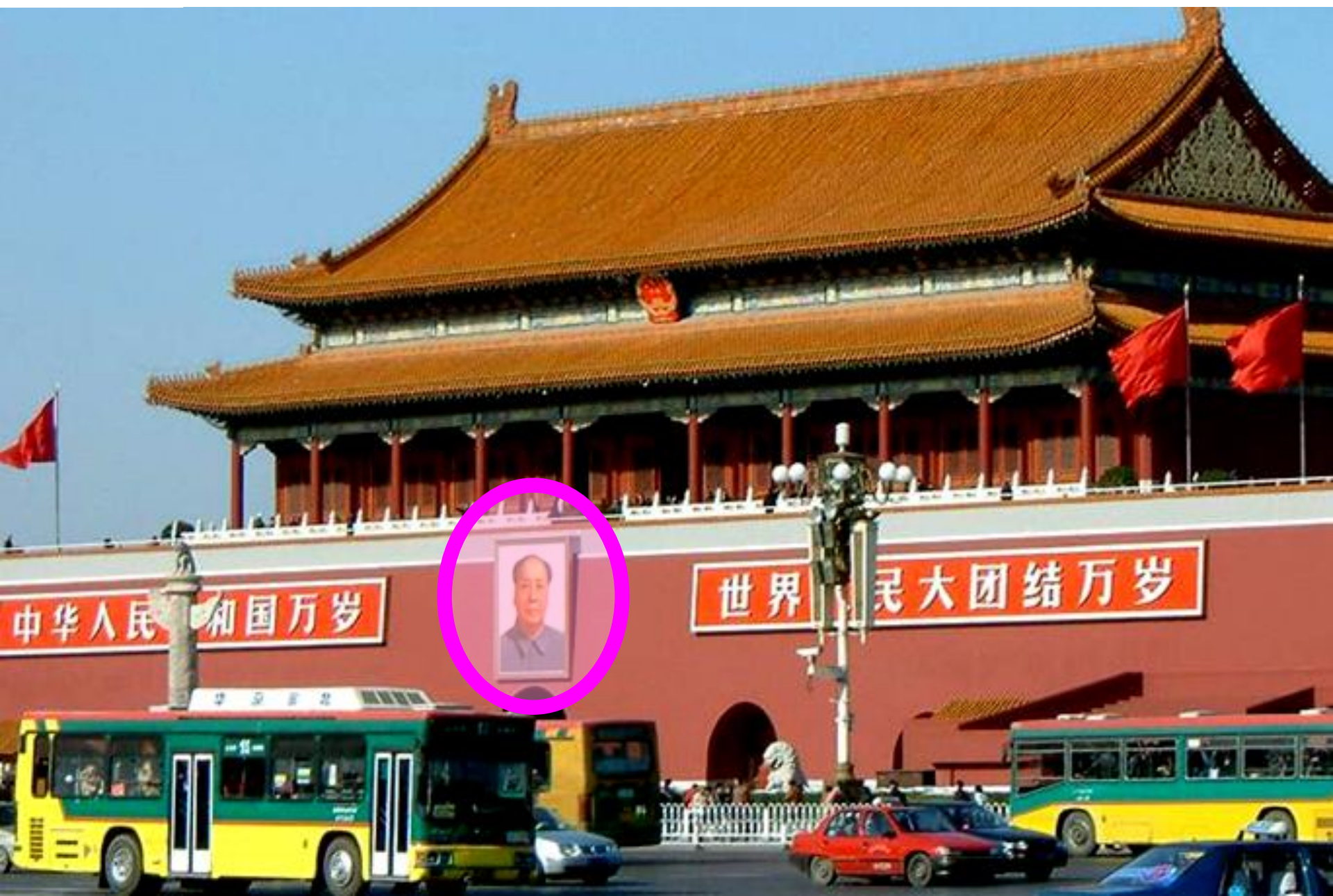
确认：这是一辆公共汽车吗？



检测:图像中有小汽车吗?



身份鉴定:是毛主席的照片吗?



目标分类

天空

建筑

旗帜

人脸

标语

墙

路灯

公汽

公汽

小车



场景及上下文分类

- 室外
- 市区
- 交通
- ...

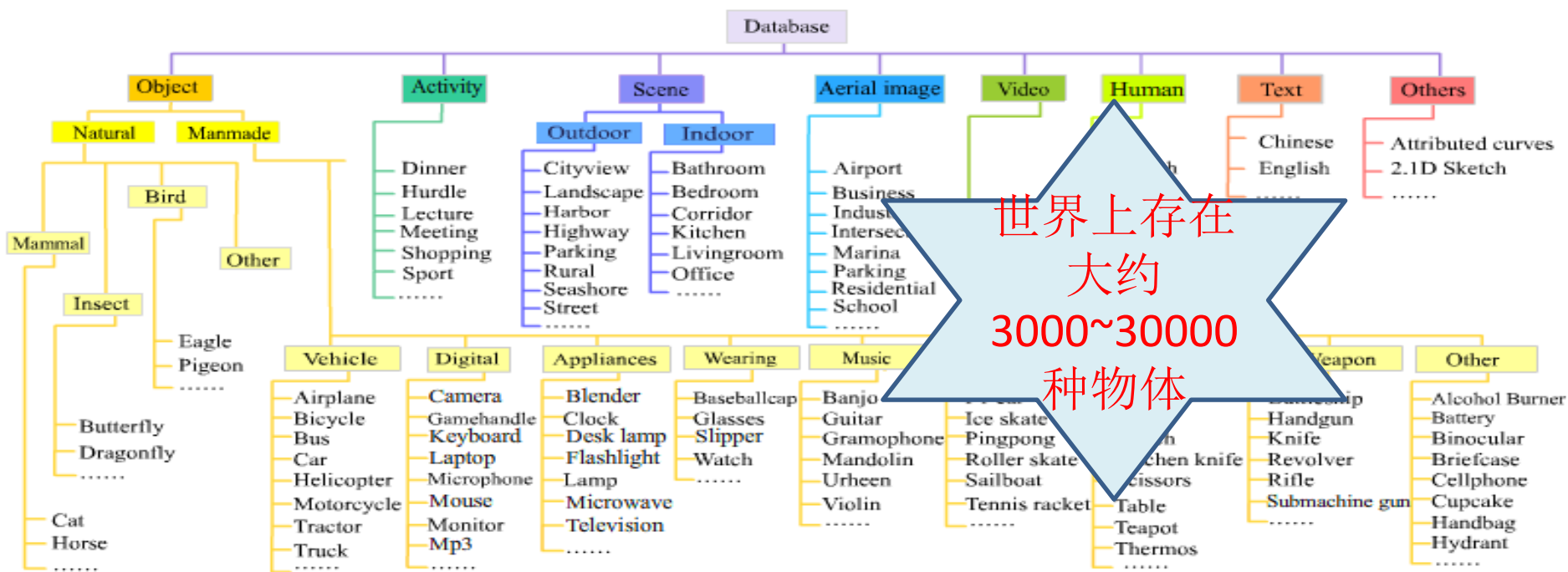




目标识别

- 目标识别面临的挑战
 - 挑战 1: 目标种类多
 - 挑战 2: 视点变化
 - 挑战 3: 亮度变化
 - 挑战 4: 遮挡
 - 挑战 5: 尺度
 - 挑战 6: 形变
 - 挑战 7: 复杂背景
 - 挑战 8: 类内变化大

挑战 1: 目标种类多



挑战 2: 视点变化



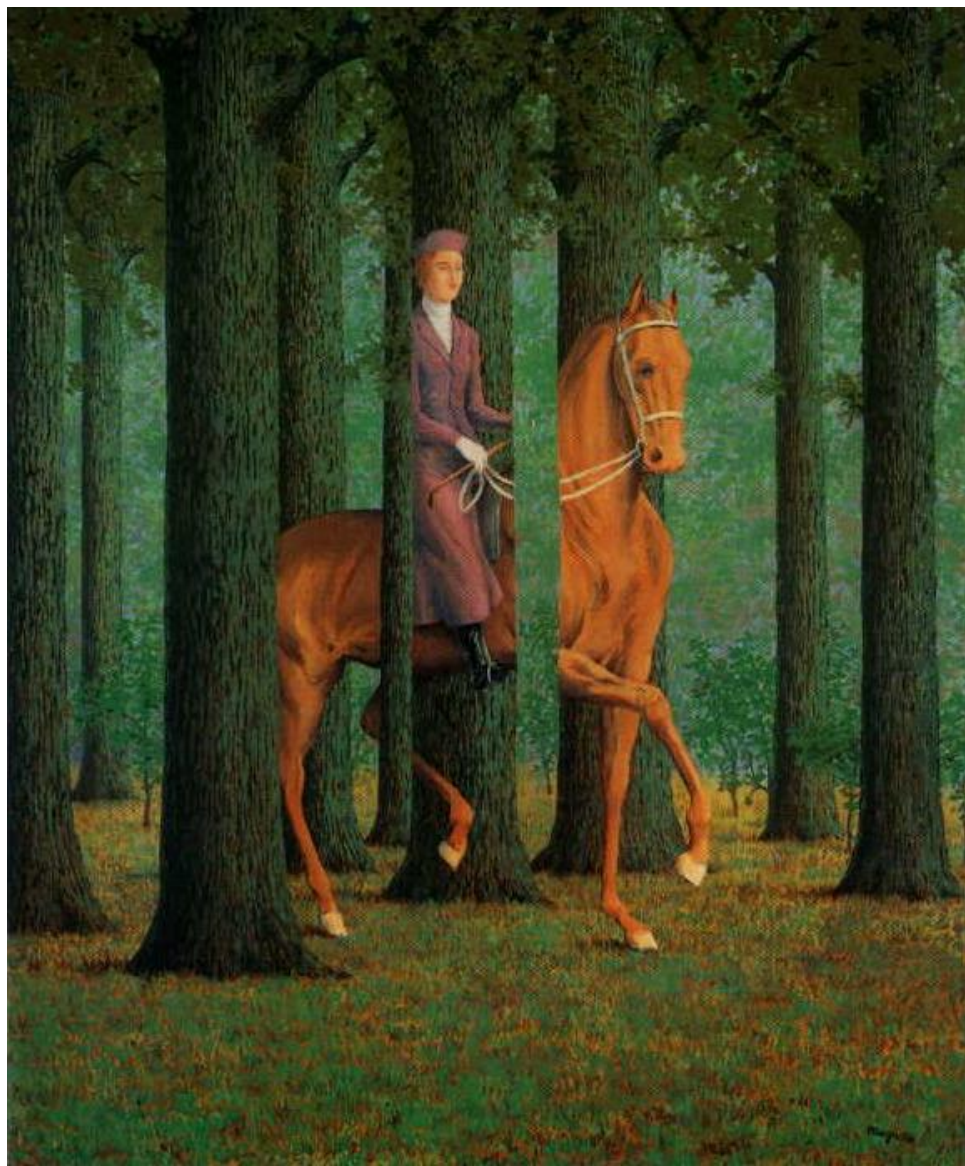


挑战 3: 亮度变化





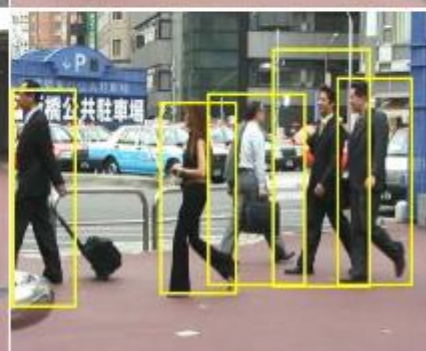
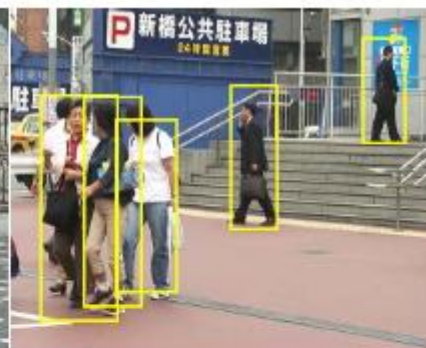
挑战 4: 遮挡



Magritte, 1957



挑战 4: 遮挡



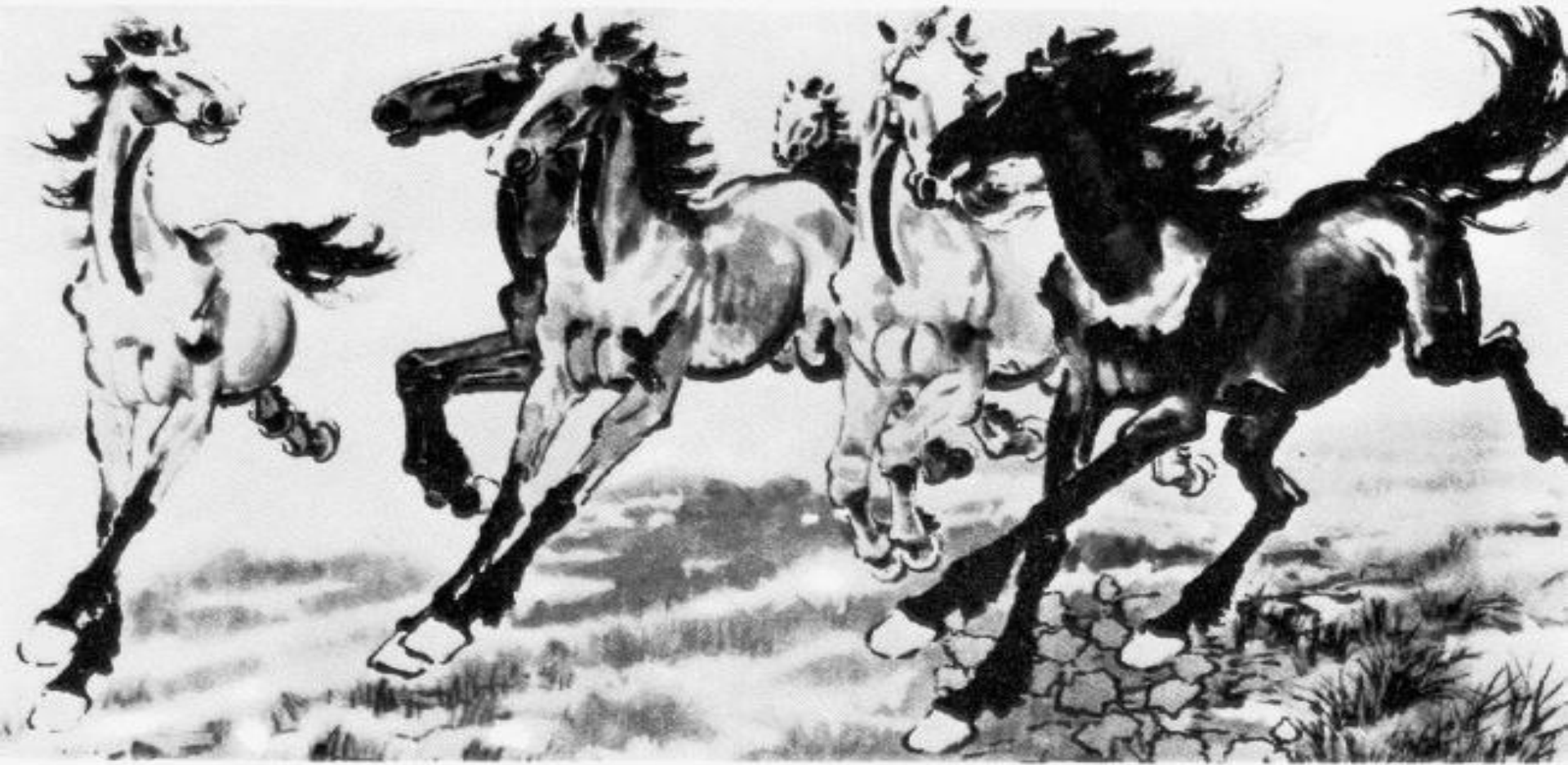


挑战 5: 尺度





挑战6: 形变



Xu, Beihong 1943



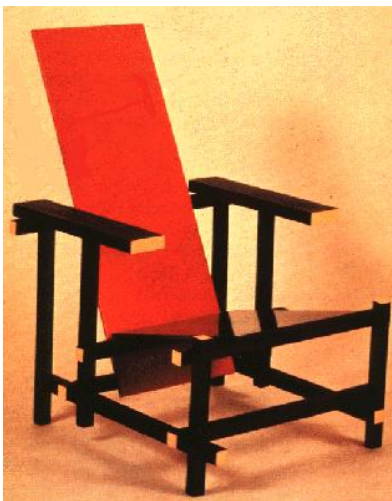
挑战 7: 复杂背景



Klimt, 1913



挑战 8: 类内变化大





目标识别方法

- 特征提取
 - Haar
 - SIFT
 - HOG
 - 直方图
 - 形状特征
 - LBP
- +
- 分类器学习
 - SVM
 - 线性判别器
 - 决策树

局部 or 全局？

PCA是不是一个
分类器？

最近邻是不是
一个分类器？



目标识别框架

- 利用一个预测函数预测类别，输入为图像的特征

$$f(\text{apple image}) = \text{"apple"}$$

$$f(\text{tomato image}) = \text{"tomato"}$$

$$f(\text{cow image}) = \text{"cow"}$$



流程示例——训练

Training
Images

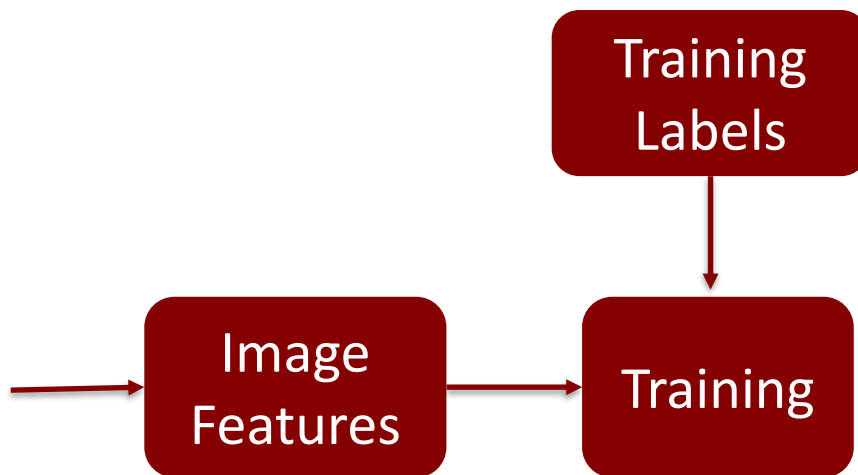


Image
Features



流程示例——训练

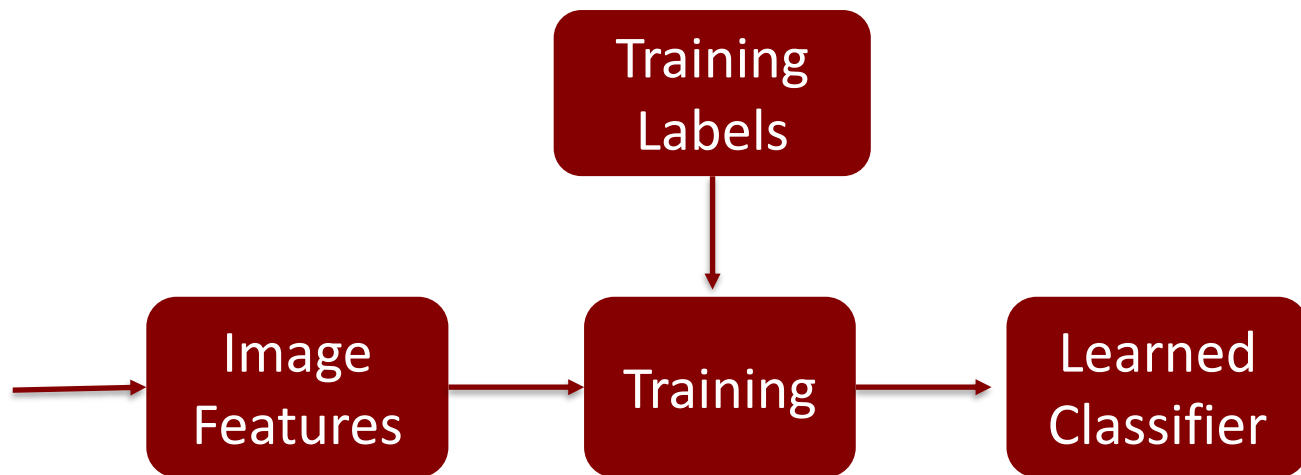
Training
Images





流程示例——训练

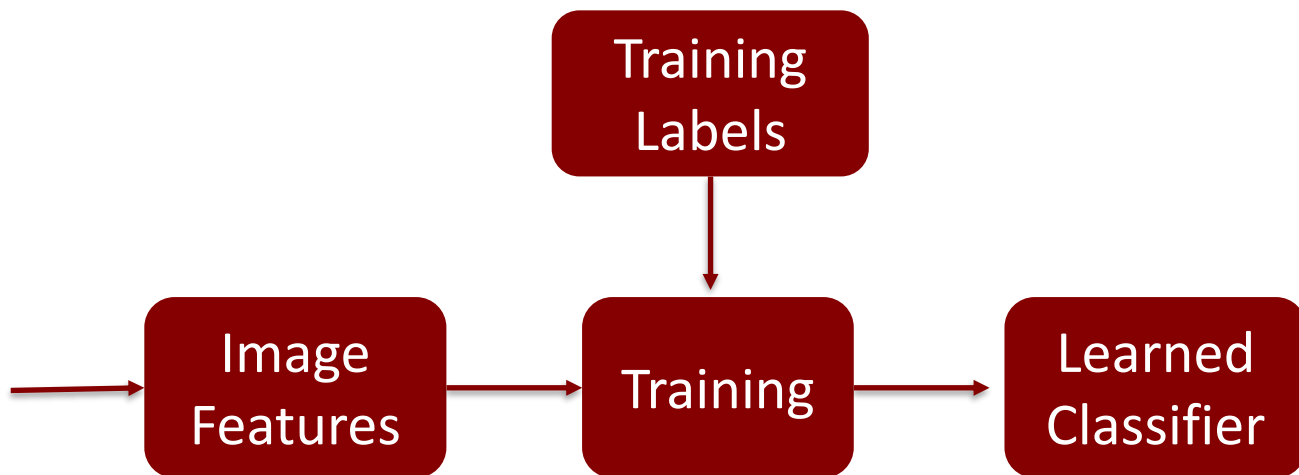
Training
Images





流程示例——训练

Training Images



Test Image



Image Features

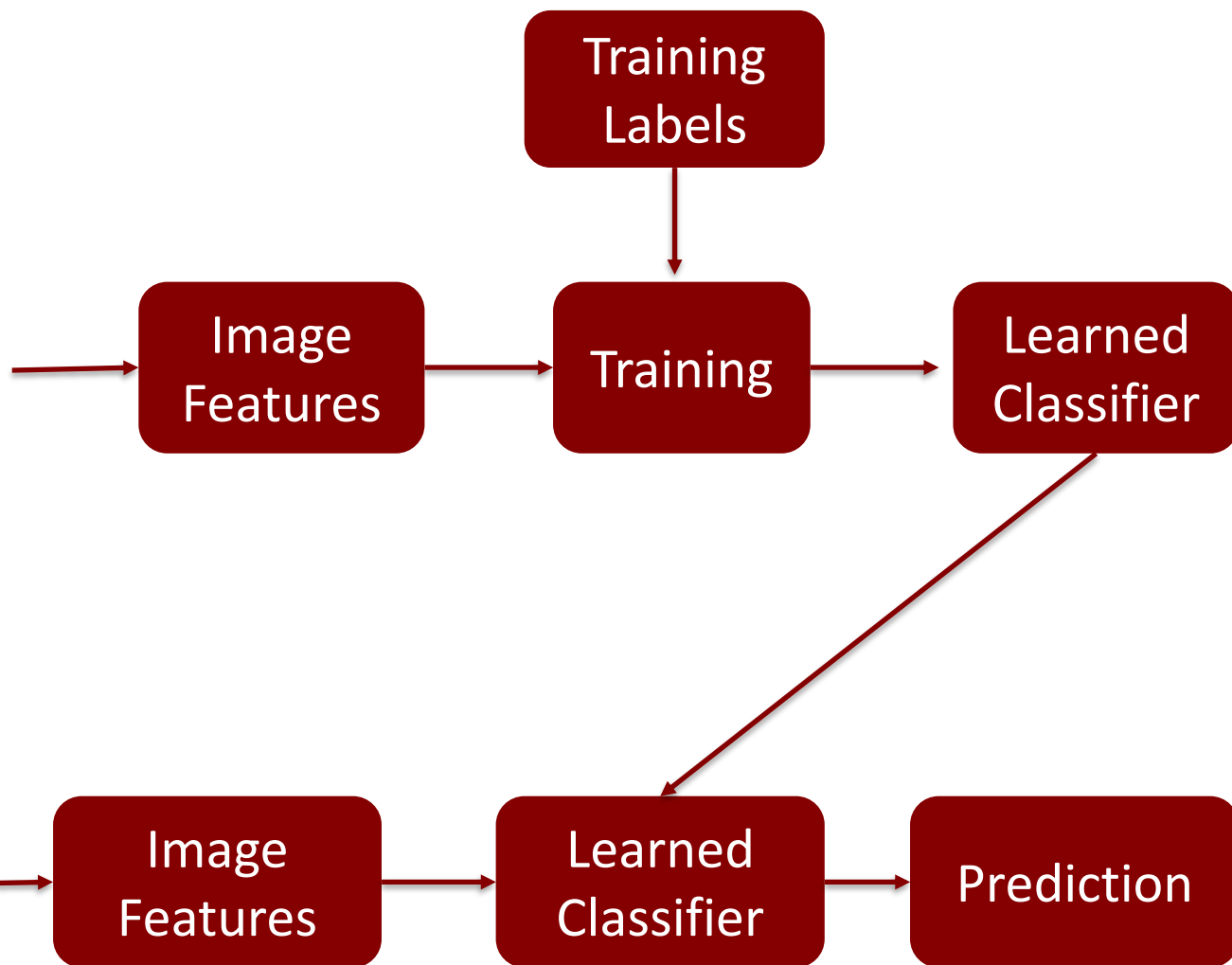


流程示例——训练

Training Images



Test Image



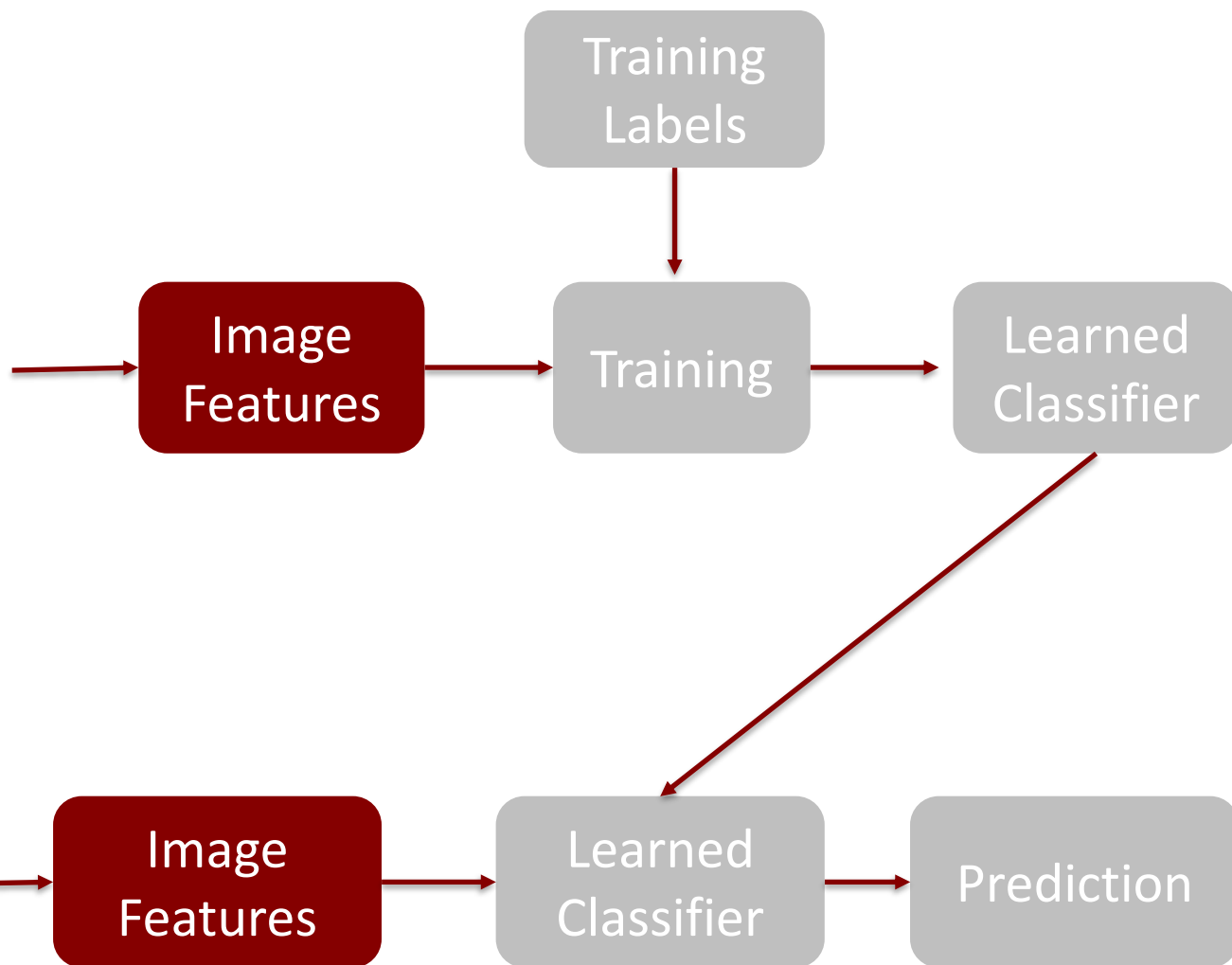


流程示例——训练

Training Images



Test Image



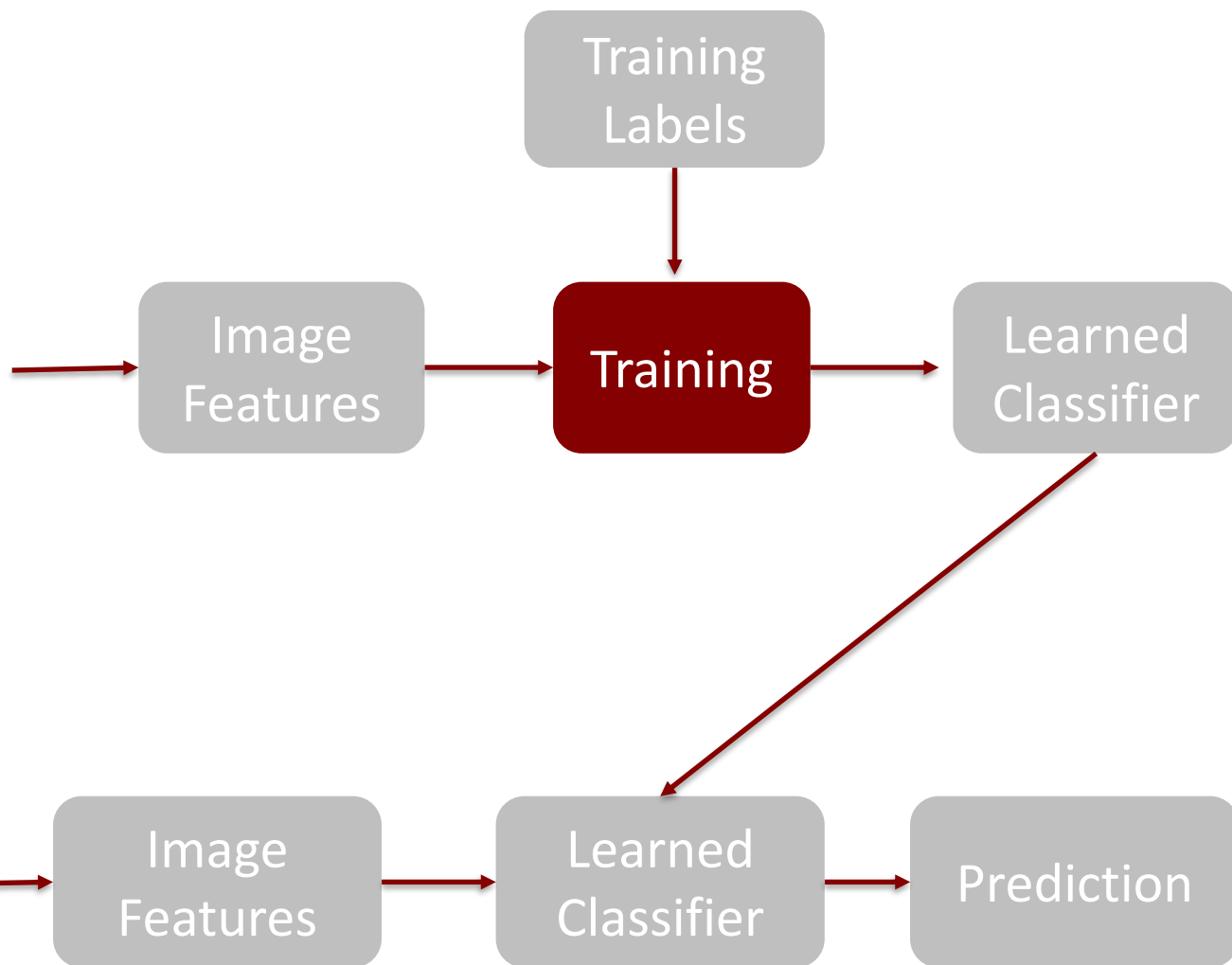


流程示例——训练

Training Images



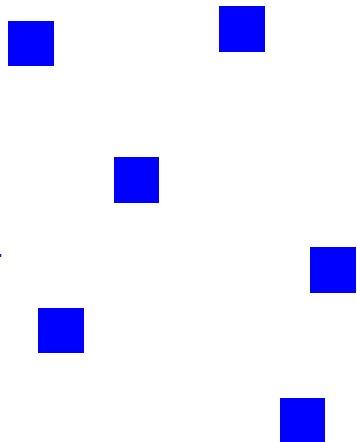
Test Image



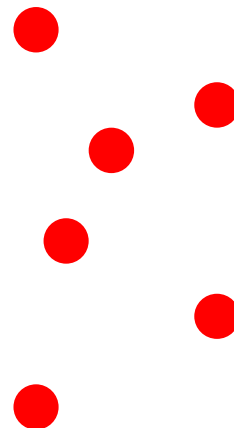


分类：最近邻分类器

Training
examples
from class 1



Training
examples
from class 2



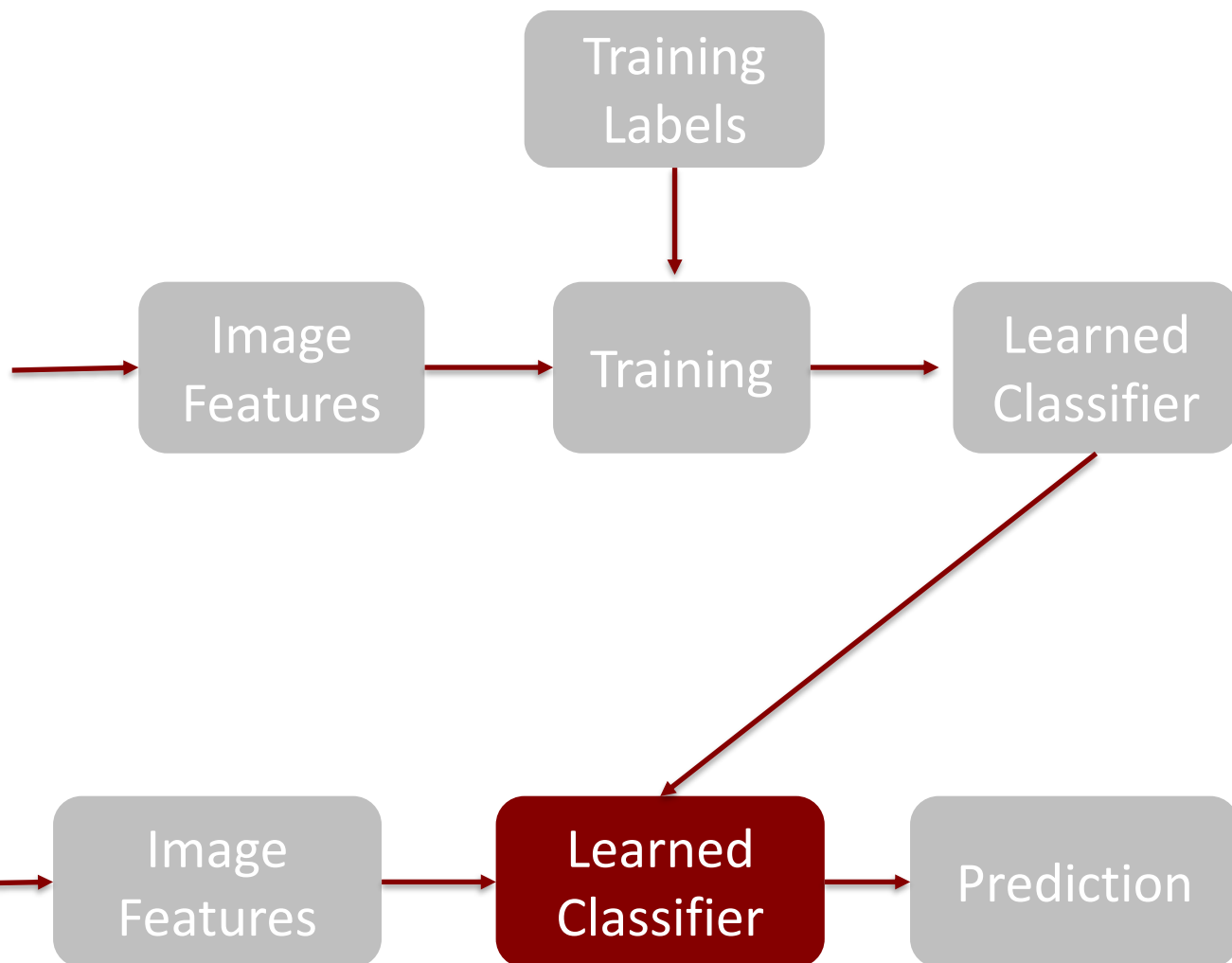


流程示例——训练

Training Images

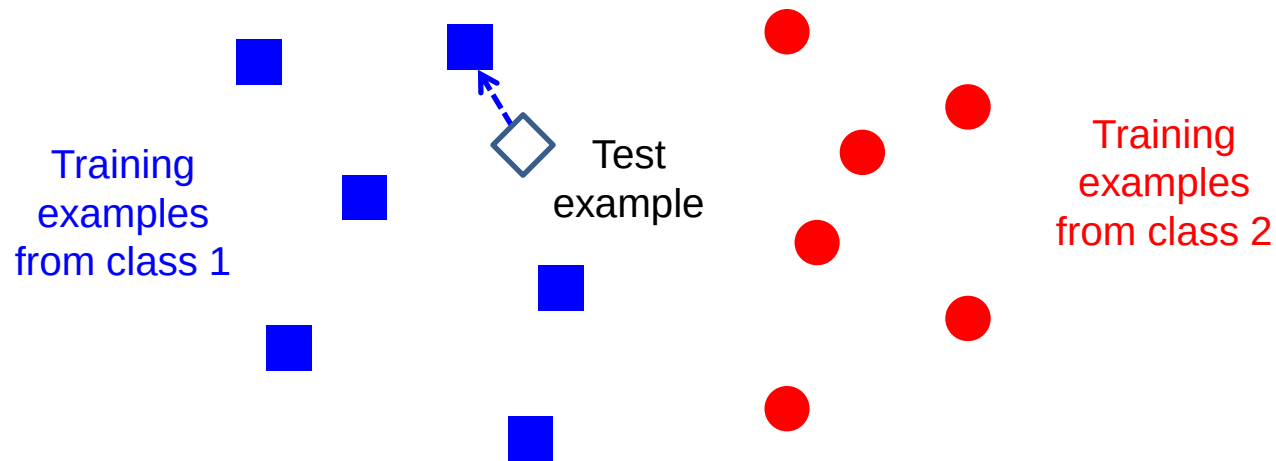


Test Image





分类：最近邻分类器





人脸检测、人脸识别

- 人脸检测是目标检测中比较成功的例子

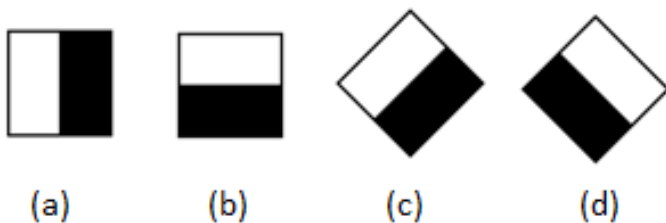




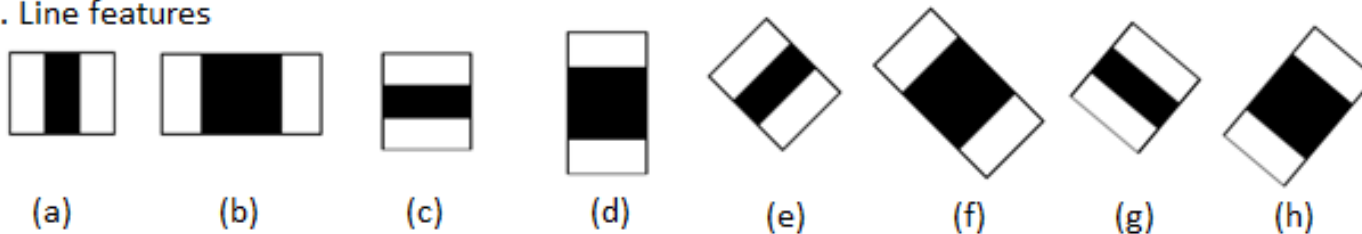
人脸检测

- Haar特征

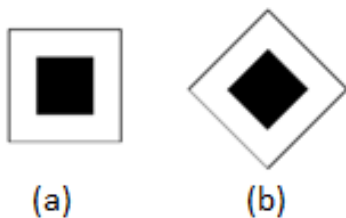
1. Edge features



2. Line features



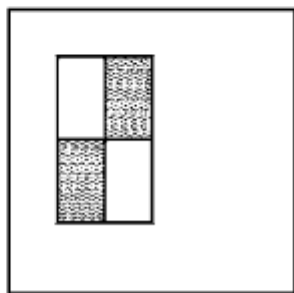
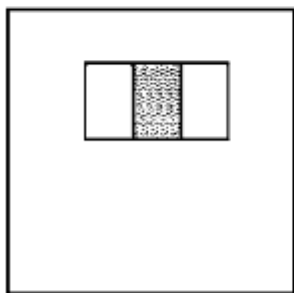
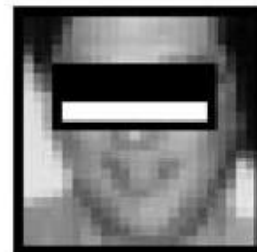
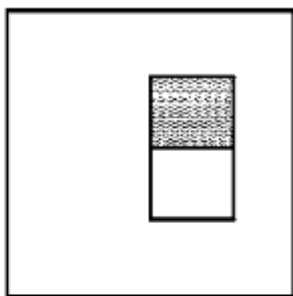
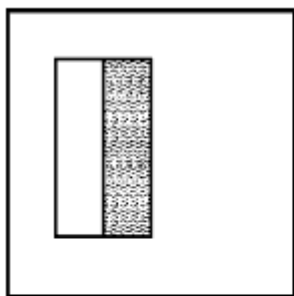
3. Center-surround features





人脸检测

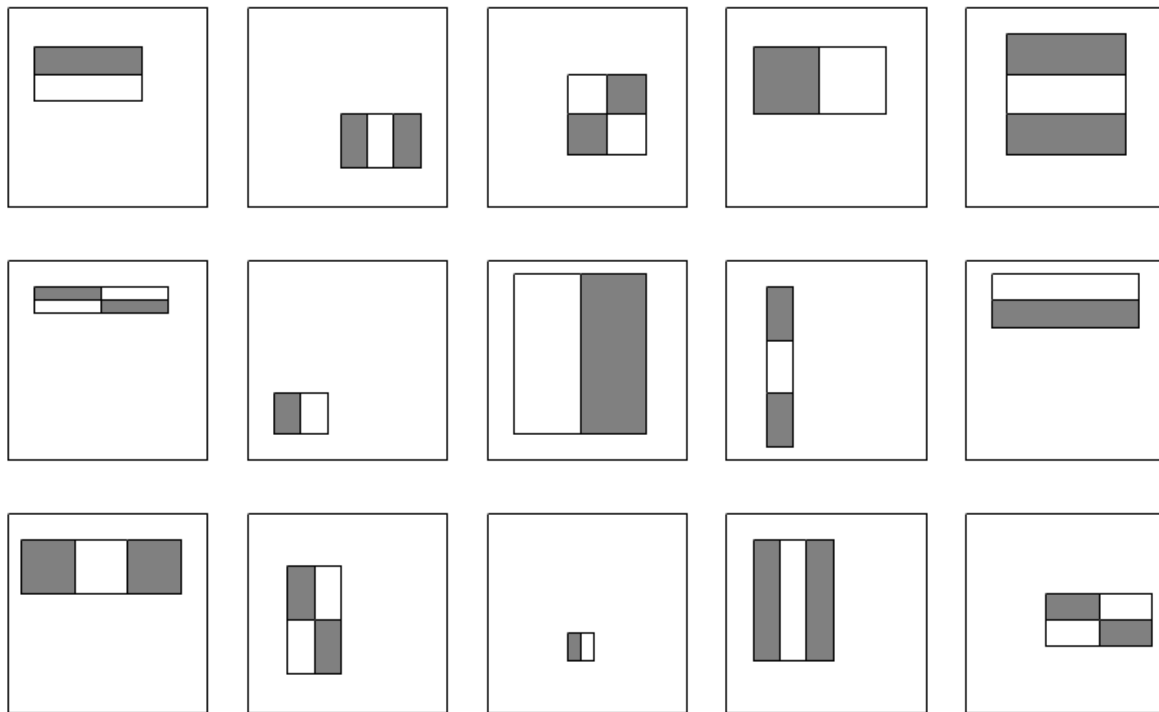
- 特征： Haar-like Features





人脸检测

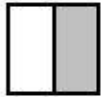
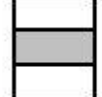
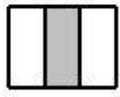
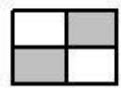
- Haar-like特征原型





人脸检测

- 过完备特征
 - 从 24×24 像素的图像上可以提取多少个Haar-like features?

特征模板	1、 	2、 	3、 	4、 	5、 
(s, t) 条件	(1, 2)	(2, 1)	(1, 3)	(3, 1)	(2, 2)



特征模板	数量
1、 	$2 \times \Omega_{(1,2)}^{24} = 2 \times \left(\left\lfloor \frac{24}{1} \right\rfloor + \left\lfloor \frac{23}{1} \right\rfloor + \cdots + \left\lfloor \frac{2}{1} \right\rfloor + 1 \right) \times \left(\left\lfloor \frac{24}{2} \right\rfloor + \left\lfloor \frac{23}{2} \right\rfloor + \cdots + \left\lfloor \frac{3}{2} \right\rfloor + 1 \right)$
2、 	$= 2 \times (24 + 23 + \cdots + 2 + 1) \times (12 + 11 + 11 + \cdots + 2 + 1 + 1)$ $= 2 \times 300 \times 144 = 2 \times 43,200$ $= 86,400$
3、 	$2 \times \Omega_{(1,3)}^{24} = 2 \times \left(\left\lfloor \frac{24}{1} \right\rfloor + \left\lfloor \frac{23}{1} \right\rfloor + \cdots + \left\lfloor \frac{2}{1} \right\rfloor + 1 \right) \times \left(\left\lfloor \frac{24}{3} \right\rfloor + \left\lfloor \frac{23}{3} \right\rfloor + \cdots + \left\lfloor \frac{4}{3} \right\rfloor + 1 \right)$
4、 	$= 2 \times (24 + 23 + \cdots + 2 + 1) \times (8 + 7 + 7 + \cdots + 1 + 1 + 1)$ $= 2 \times 300 \times 92 = 2 \times 27,600$ $= 55,200$
5、 	$\Omega_{(2,2)}^{24} = \left(\left\lfloor \frac{24}{2} \right\rfloor + \left\lfloor \frac{23}{2} \right\rfloor + \cdots + \left\lfloor \frac{3}{2} \right\rfloor + 1 \right) \times \left(\left\lfloor \frac{24}{2} \right\rfloor + \left\lfloor \frac{23}{2} \right\rfloor + \cdots + \left\lfloor \frac{3}{2} \right\rfloor + 1 \right)$ $= (12 + 11 + 11 + \cdots + 2 + 1 + 1) \times (12 + 11 + 11 + \cdots + 2 + 1 + 1)$ $= 144 \times 144$ $= 20,736$
总数	$\Omega^{24} = 162,336$



人脸检测

- 过完备特征
 - 从 24×24 像素的图像上可以提取多少个Haar-like features?
 - 160,000!

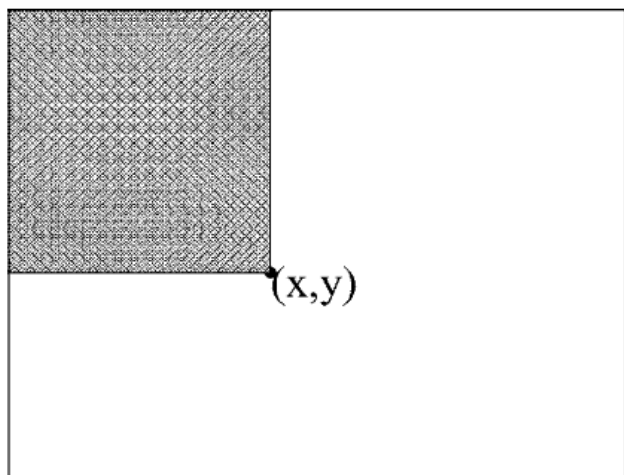
窗口大小	36×36	30×30	24×24	20×20	16×16
特征数量	816,264	394,725	162,336	78,460	32,384

- 运算量过大



人脸检测

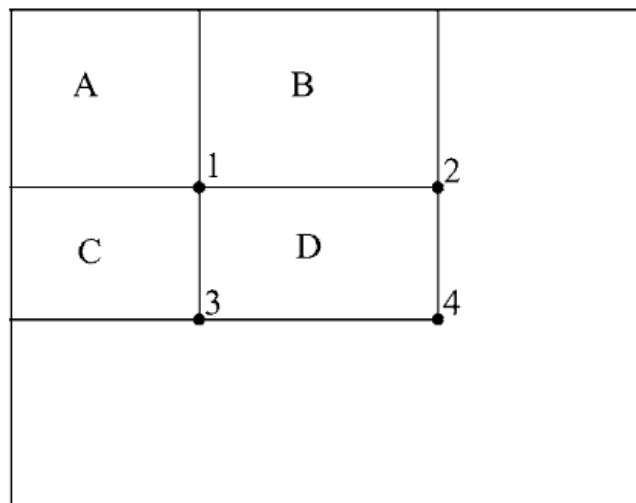
- 积分图像



$$ii(x, y) = \sum_{x' \leq x, y' \leq y} i(x', y')$$

$$s(x, y) = s(x, y - 1) + i(x, y)$$

$$ii(x, y) = ii(x - 1, y) + s(x, y)$$



$$\begin{aligned} D &= 1 + 4 - (2 + 3) \\ &= A + (A + B + C + D) - (A + C + A + B) \\ &= D \end{aligned}$$



人脸检测

- 分类器
 - **AdaBoost (Adaptive Boosting) 分类器**
 - **Boosting:** 是一种提高任意给定学习算法准确度的方法。这种方法依次顺序地加入多个分量分类器(弱分类器即可)，其中每个分量分类器的训练集都选自己有的其他各个分类器所给出的“最富信息(most informative)”的样本点组成，而最终的判决结果则是根据这些分量分类器的结果共同决定。
 - “三个臭皮匠，赛过诸葛亮”



人脸检测

- 分类器
 - **AdaBoost (Adaptive Boosting) 分类器**
 - 允许设计者不断地加入新的弱分类器，直到达到某个预定的足够小的错误率。
 - 每个训练样本都被赋予一个权重，表明它被某个分量分类器选中的概率。如果每个样本点已经被准确地分类，那么在构造下一个训练集的时候，它被选中的概率就降低；相反，如果某个样本点没有被正确地分类，那么它的权重得到提高。



人脸检测

- 分类器
 - AdaBoost (Adaptive Boosting) 分类器

Given: $(x_1, y_1), \dots, (x_m, y_m)$ where $x_i \in X, y_i \in Y = \{-1, +1\}$

Initialize $D_1(i) = 1/m$.

For $t = 1, \dots, T$:

- Train weak learner using distribution D_t .
- Get weak hypothesis $h_t : X \rightarrow \{-1, +1\}$ with error

$$\epsilon_t = \Pr_{i \sim D_t} [h_t(x_i) \neq y_i].$$

- Choose $\alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$.
- Update:

$$\begin{aligned} D_{t+1}(i) &= \frac{D_t(i)}{Z_t} \times \begin{cases} e^{-\alpha_t} & \text{if } h_t(x_i) = y_i \\ e^{\alpha_t} & \text{if } h_t(x_i) \neq y_i \end{cases} \\ &= \frac{D_t(i) \exp(-\alpha_t y_i h_t(x_i))}{Z_t} \end{aligned}$$

where Z_t is a normalization factor (chosen so that D_{t+1} will be a distribution).

Output the final hypothesis:

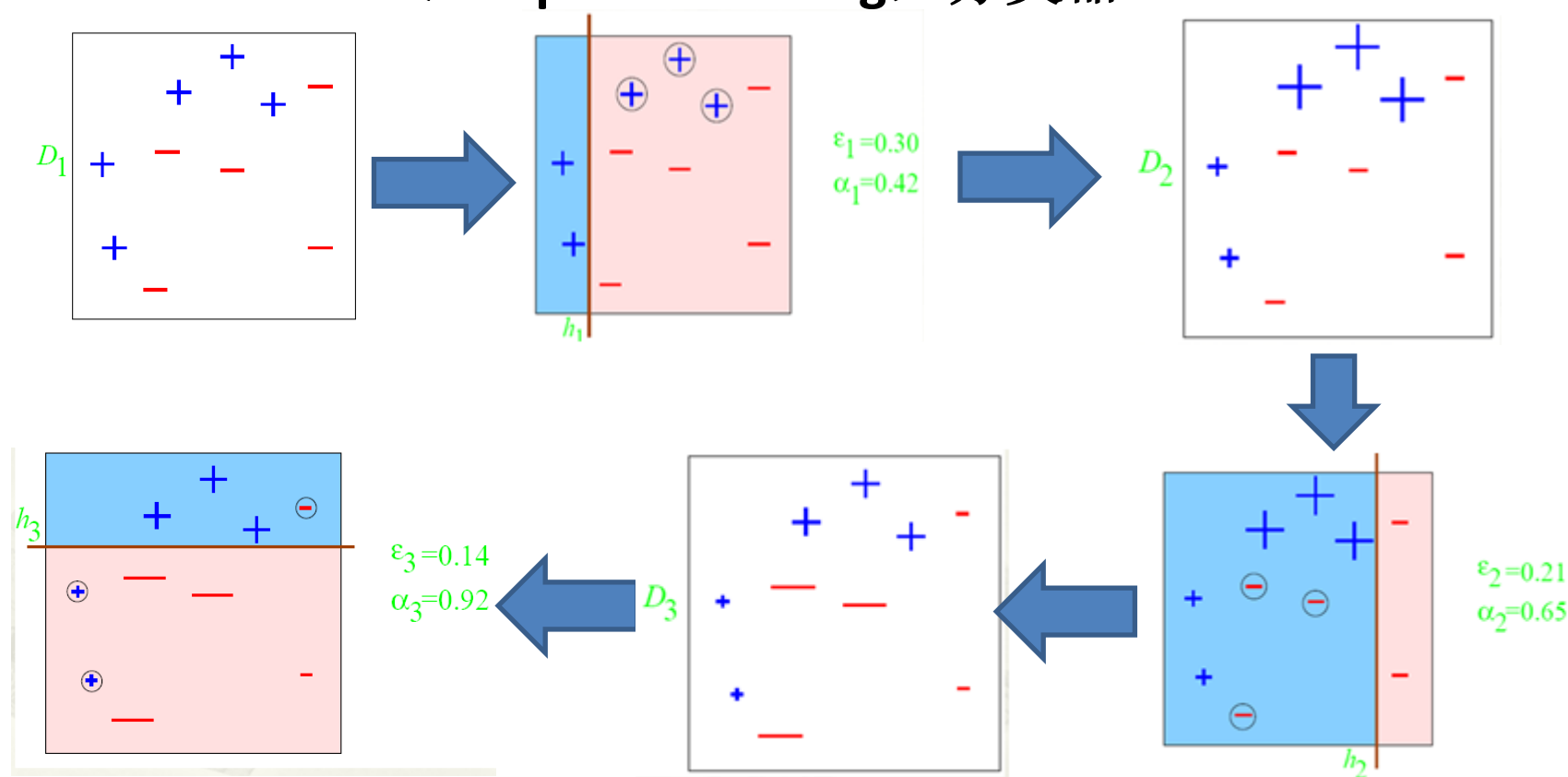
$$H(x) = \text{sign} \left(\sum_{t=1}^T \alpha_t h_t(x) \right).$$



人脸检测

- 分类器

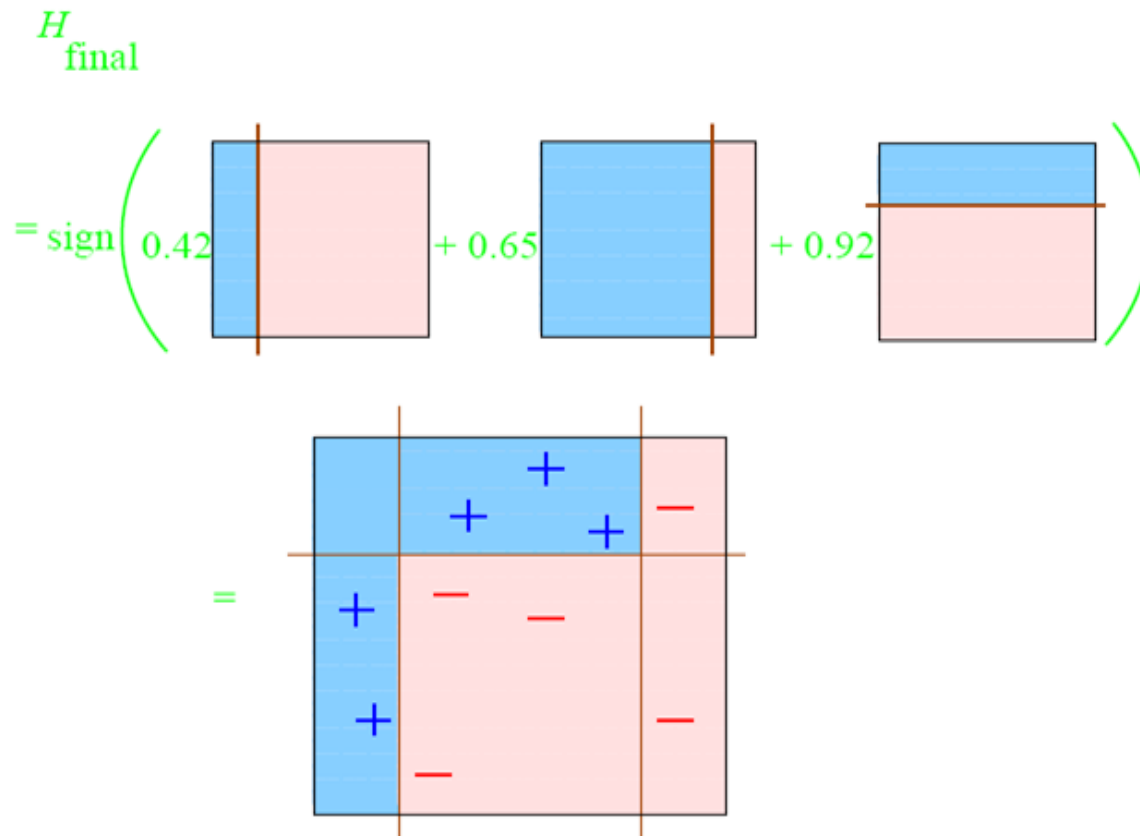
- AdaBoost (Adaptive Boosting) 分类器





人脸检测

- 分类器
 - AdaBoost (Adaptive Boosting) 分类器





人脸检测

- **AdaBoost (Adaptive Boosting) 分类器**

- 弱分类器

- 决策桩(stumps)

$$h(x, f, p, \theta) = \begin{cases} 1 & \text{if } pf(x) > p\theta \\ -1 & \text{otherwise} \end{cases}$$

- 每一个弱分类器由一个特征组成

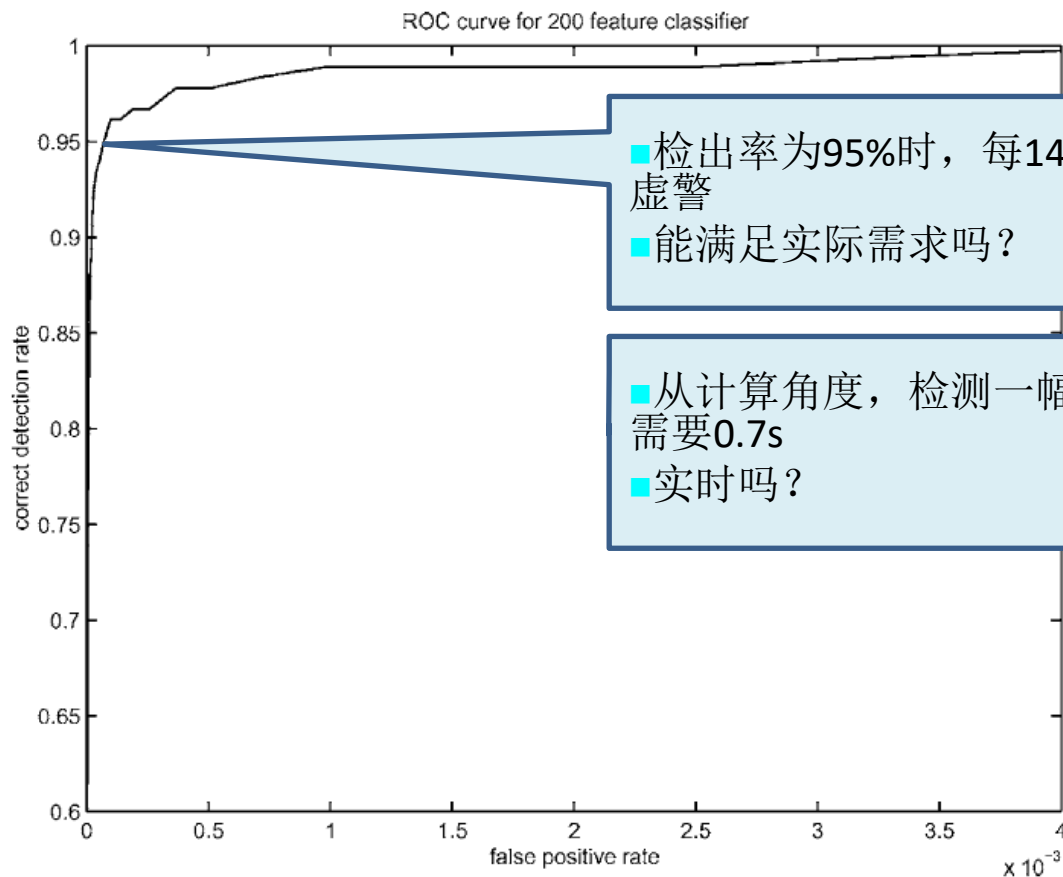
- 正负样本不平衡时:

- 初始化权重: $D_{1,i} = \frac{1}{2m}, \frac{1}{2n}$, m, n 分别为正样本、负样本数量



人脸检测

- 分类器
 - 分类结果



■ 检出率为95%时，每14084次检测有1个虚警

■ 能满足实际需求吗？

■ 从计算角度，检测一幅384*288的图像需要0.7s

■ 实时吗？



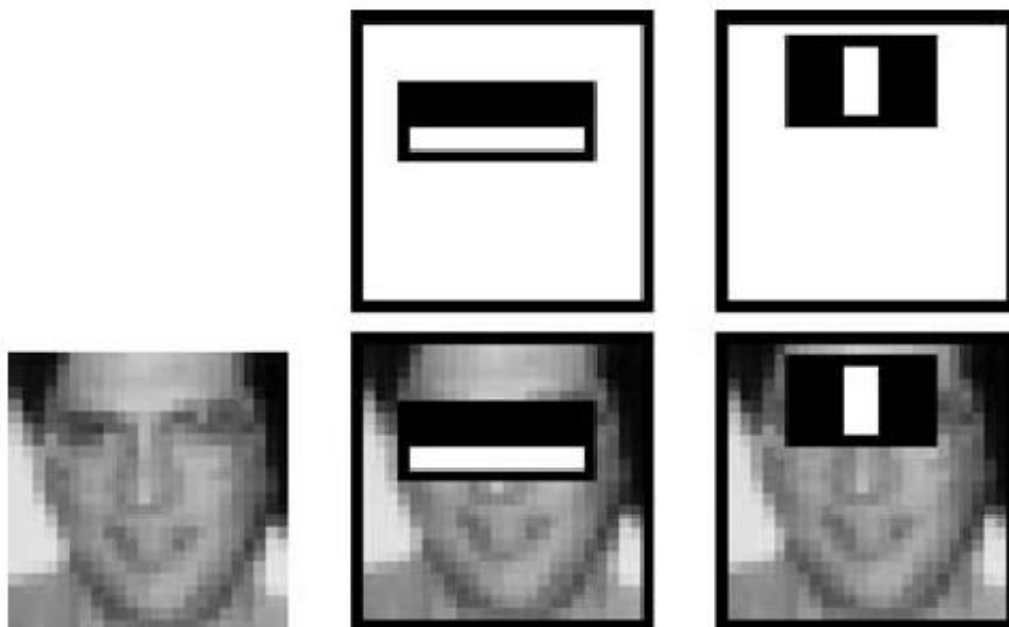
人脸检测

- 层级式(Cascade)分类器
 - 对于一幅图像，只有极少量扫描窗中存在人脸。也就是说大多数扫描窗中都是非人脸。
 - 使用少量的特征的组成的分类器排除掉大部分非人脸的区域。
 - “先易后难”



人脸检测

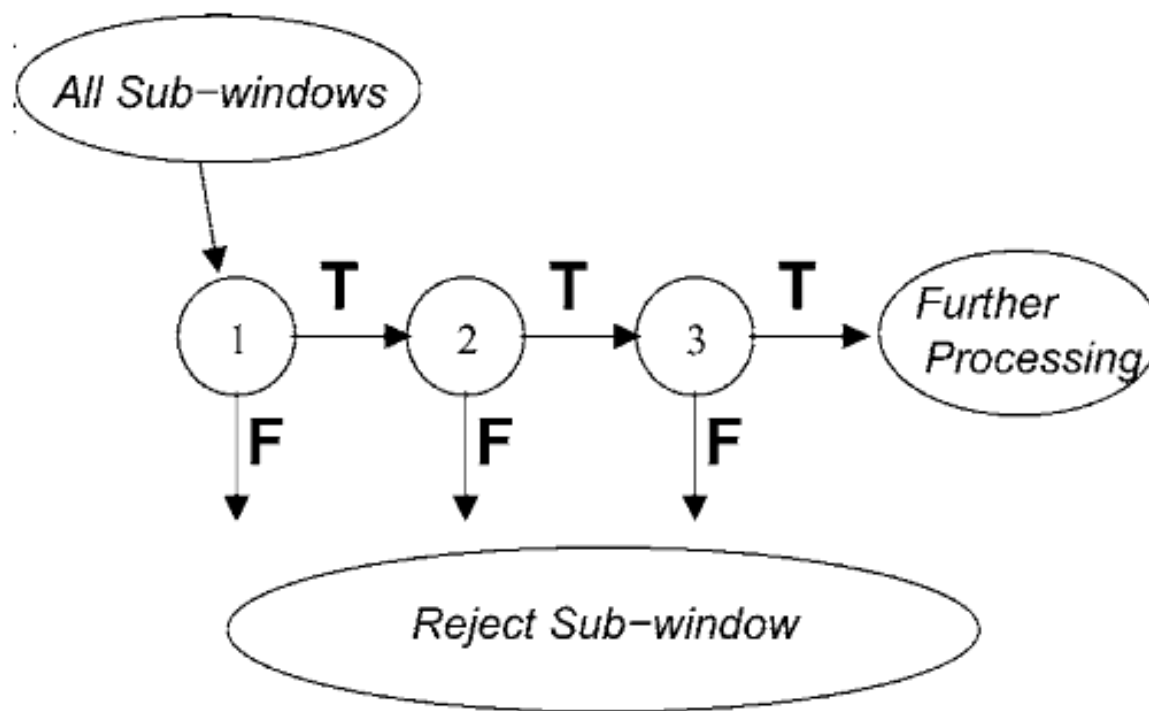
- 层级式(Cascade)分类器
 - 先易后难





人脸检测

- 层级式(Cascade)分类器





人脸检测

- 层级式(Cascade)分类器

- 检测率和虚警率

- False Positive Rate (假阳率)

$$F = \prod_{i=1}^K f_i,$$

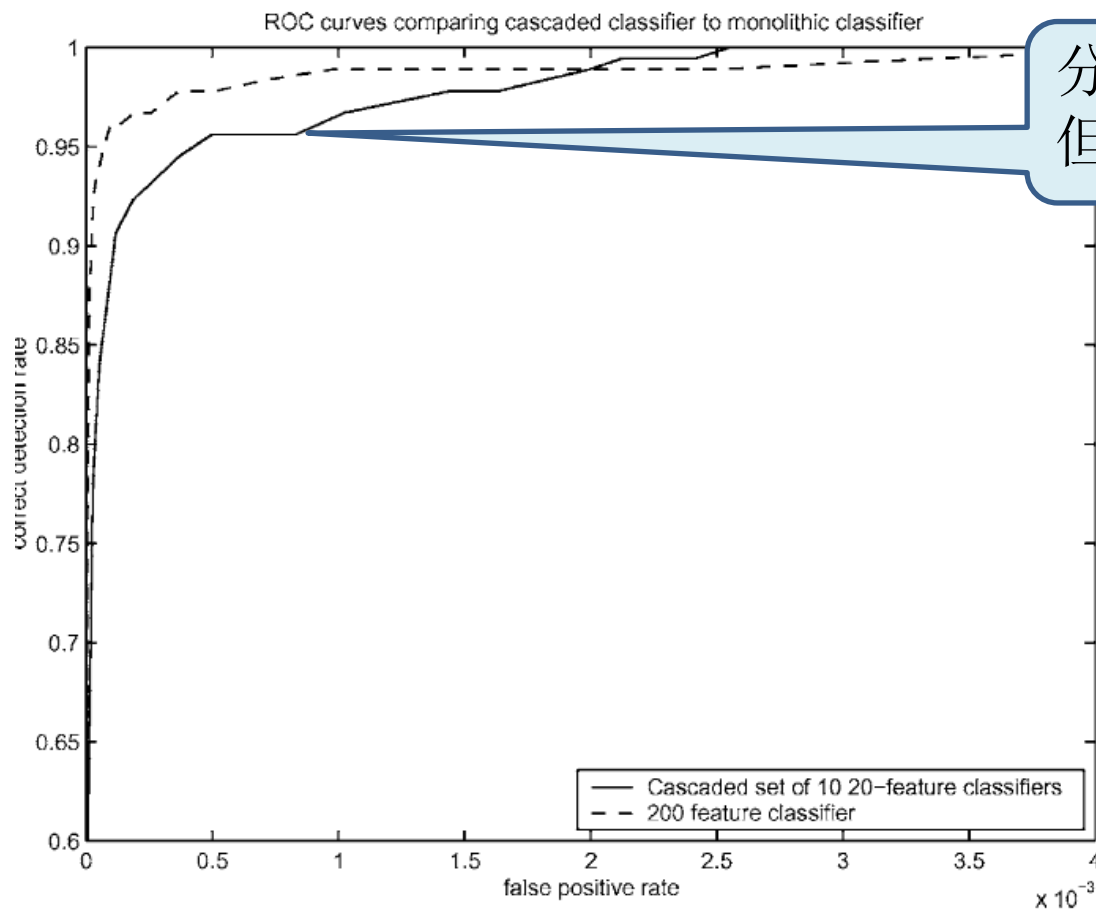
- Detection Rate

$$D = \prod_{i=1}^K d_i,$$



人脸检测

- 层级式(Cascade)分类器



分类效果有些下降
但速度快了10倍！



人脸检测

- 层级式(Cascade)分类器
 - 一般来说，使用较多的特征会得到检测率高、虚警率低的分类器。
 - 使用的特征越多，计算耗时越多
 - 如何折中？



人脸检测

- 层级式(Cascade)分类器

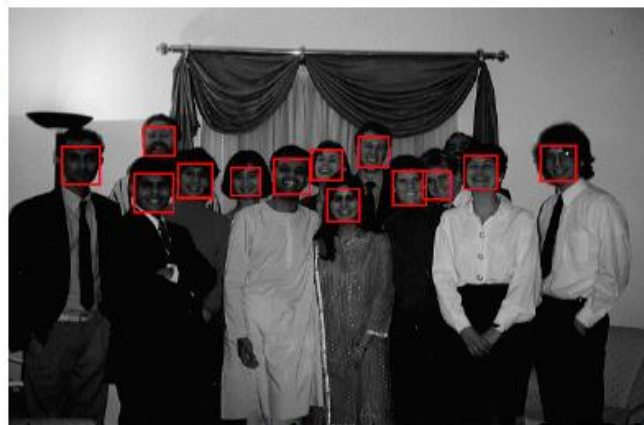
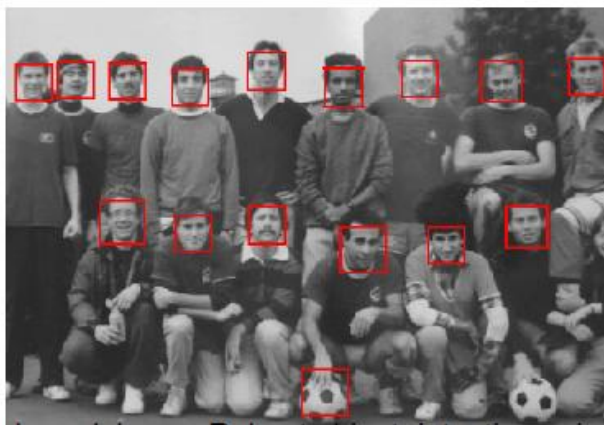
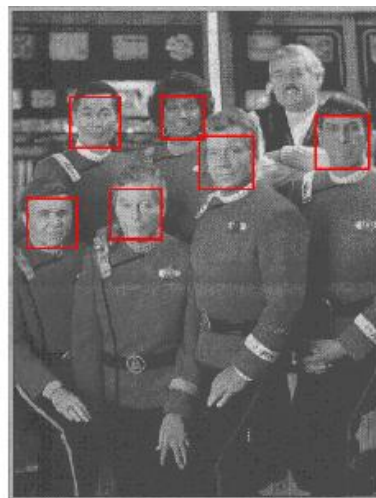
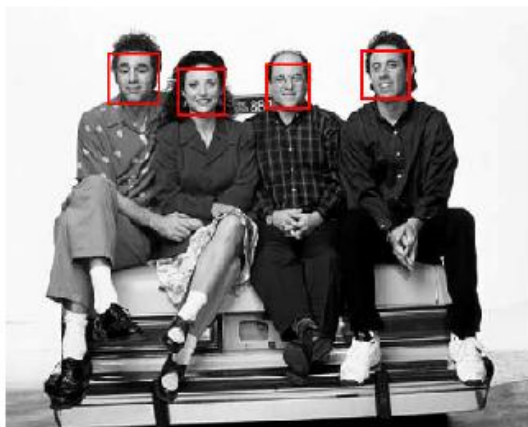
- User selects values for f , the maximum acceptable false positive rate per layer and d , the minimum acceptable detection rate per layer.
- User selects target overall false positive rate, F_{target} .
- P = set of positive examples
- N = set of negative examples
- $F_0 = 1.0$; $D_0 = 1.0$
- $i = 0$
- while $F_i > F_{target}$
 - $i \leftarrow i + 1$
 - $n_i = 0$; $F_i = F_{i-1}$
 - while $F_i > f \times F_{i-1}$
 - * $n_i \leftarrow n_i + 1$
 - * Use P and N to train a classifier with n_i features using AdaBoost
 - * Evaluate current cascaded classifier on validation set to determine F_i and D_i .
 - * Decrease threshold for the i th classifier until the current cascaded classifier has a detection rate of at least $d \times D_{i-1}$ (this also affects F_i)
 - $N \leftarrow \emptyset$
 - If $F_i > F_{target}$ then evaluate the current cascaded detector on the set of non-face images and put any false detections into the set N



人脸检测

- 人脸检测结果

窗口扫描

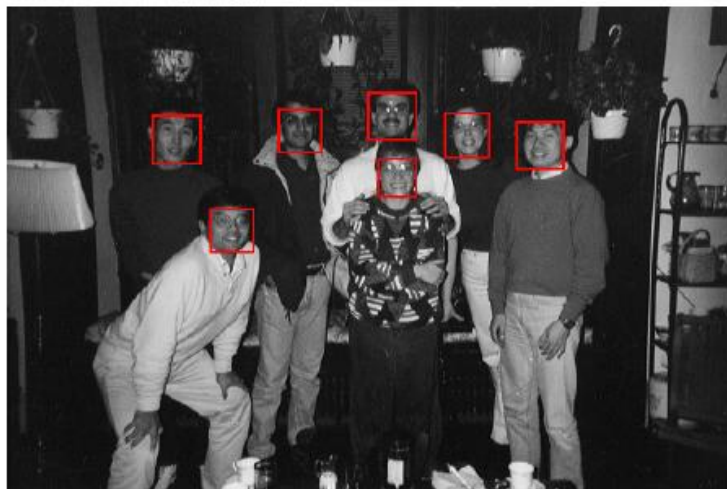
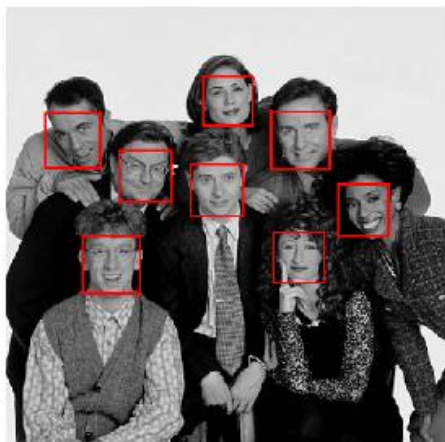
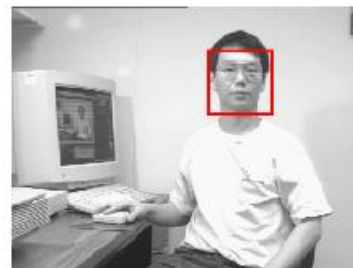
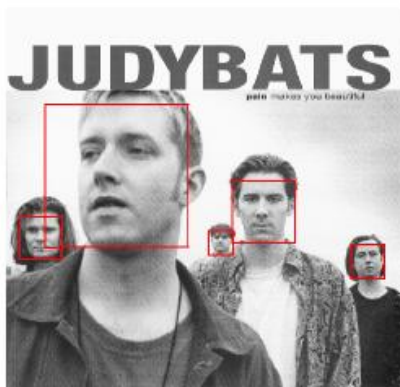




人脸检测

- 人脸检测结果

多尺度搜索, NMS





人脸检测

- 算法贡献总结
 - 特征（Haar-like, 过完备特征集）
 - 积分图像
 - 基于AdaBoost的特征选择方法。选择弱分类器即选择特征
 - 层级式分类器



人脸检测



06.27.08

Disney • PIXAR



Click to Enlarge

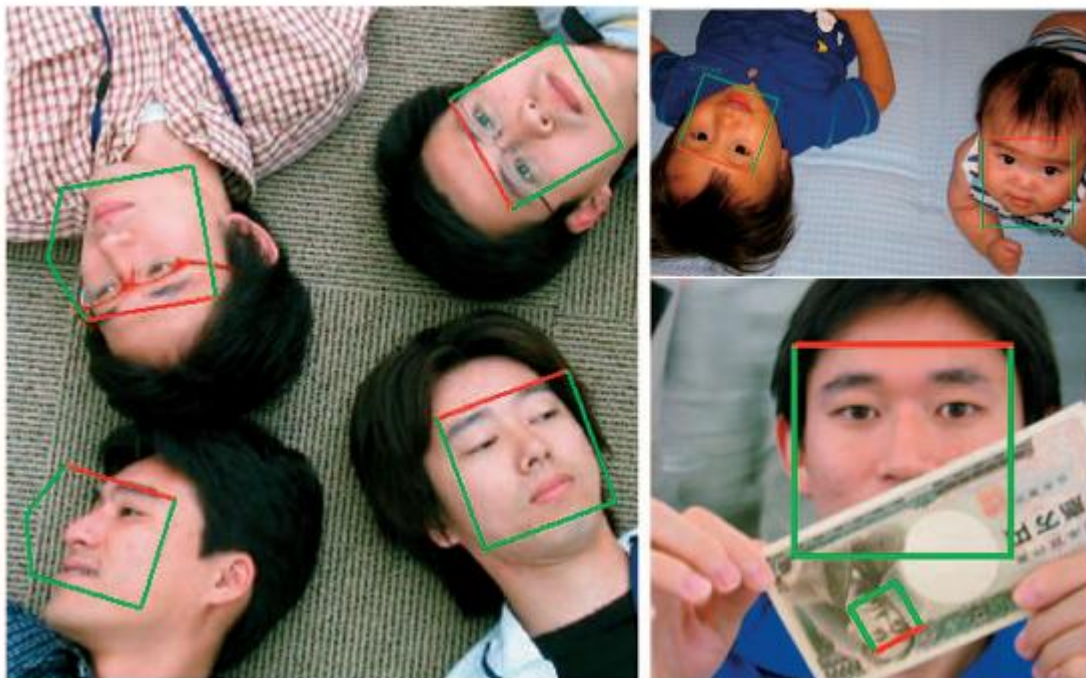


人脸检测





人脸检测



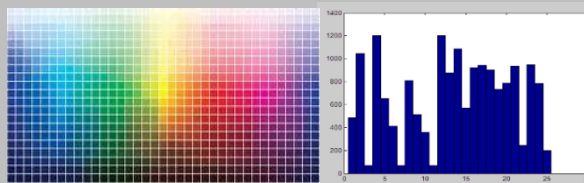


目标检测识别中的几个问题——图像特征

Input image



Color: Quantize RGB values



Invariance?

- ? Translation
- ? Scale
- ? Rotation
- ? Occlusion

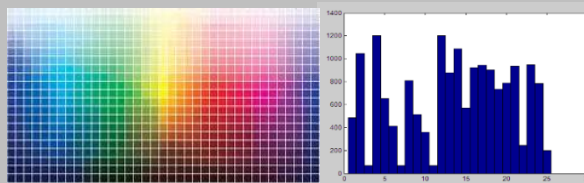


目标检测识别中的几个问题——图像特征

Input image



Color: Quantize RGB values

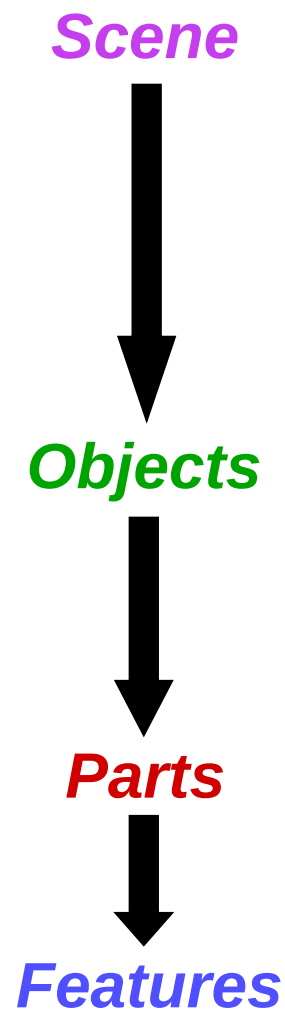
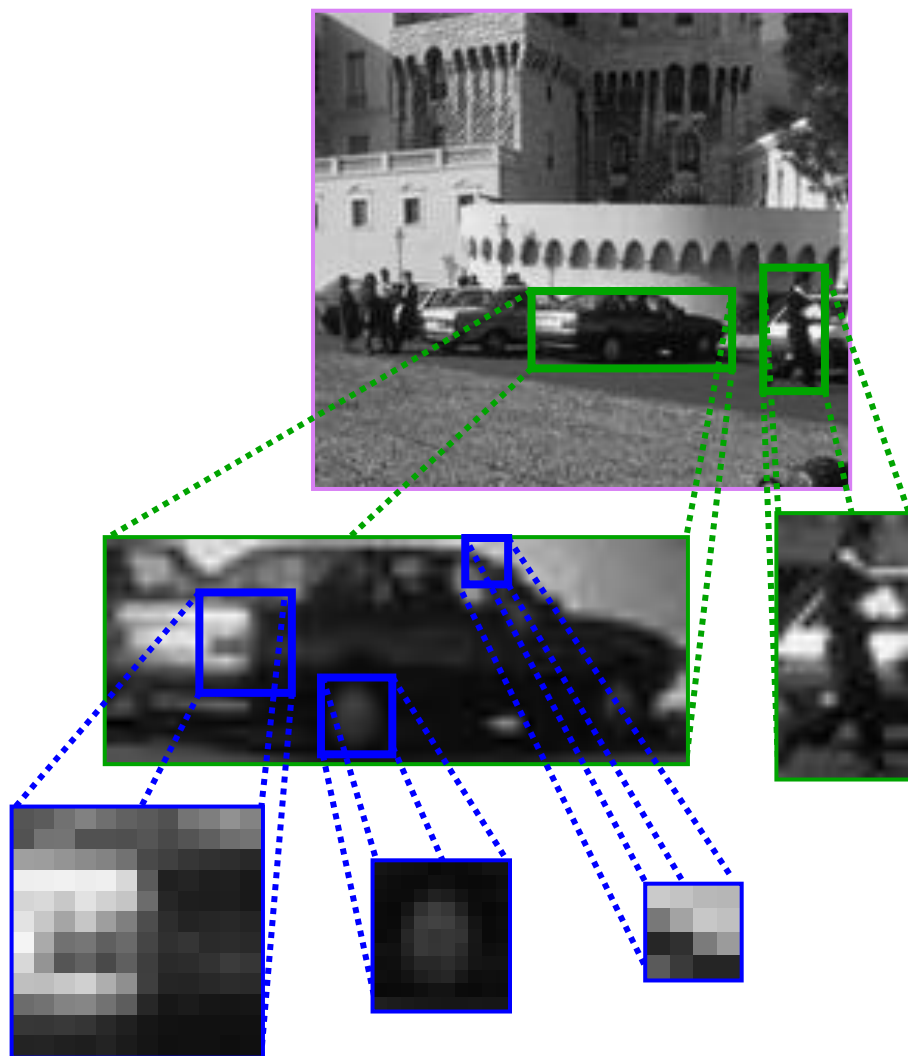


Invariance?

- 😊 Translation
- ? Scale
- 😊 Rotation (in-planar)
- 😞 Occlusion



目标检测识别中的几个问题——层次化分类





目标检测识别中的几个问题——判别式与生成式



$$p(\text{zebra} | \text{image})$$

vs.

$$p(\text{no zebra} | \text{image})$$

■ Bayes rule:

$$\underbrace{\frac{p(\text{zebra} | \text{image})}{p(\text{no zebra} | \text{image})}}_{\text{posterior ratio}} = \underbrace{\frac{p(\text{image} | \text{zebra})}{p(\text{image} | \text{no zebra})}}_{\text{likelihood ratio}} \cdot \underbrace{\frac{p(\text{zebra})}{p(\text{no zebra})}}_{\text{prior ratio}}$$



目标检测识别中的几个问题——判别式与生成式

$$\underbrace{\frac{p(\text{zebra} | \text{image})}{p(\text{no zebra} | \text{image})}}_{\text{posterior ratio}} = \underbrace{\frac{p(\text{image} | \text{zebra})}{p(\text{image} | \text{no zebra})}}_{\text{likelihood ratio}} \cdot \underbrace{\frac{p(\text{zebra})}{p(\text{no zebra})}}_{\text{prior ratio}}$$

- 判别式对后验概率建模
- 生成式对似然概率和先验概率建模



目标检测识别中的几个问题——判别式

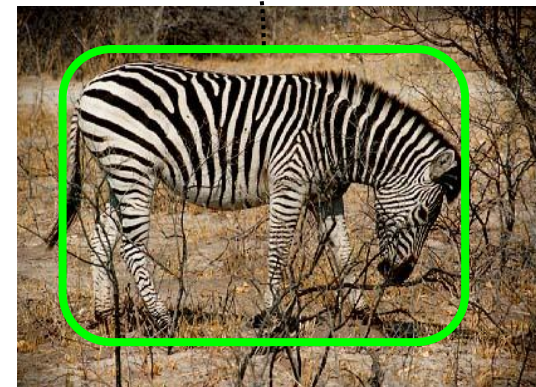
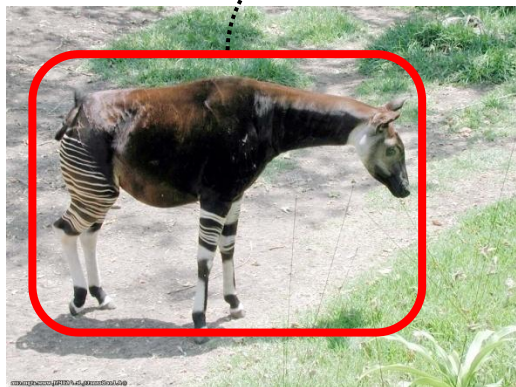
■ 直接建模

$$\frac{p(\text{zebra} | \text{image})}{p(\text{no zebra} | \text{image})}$$

Decision
boundary

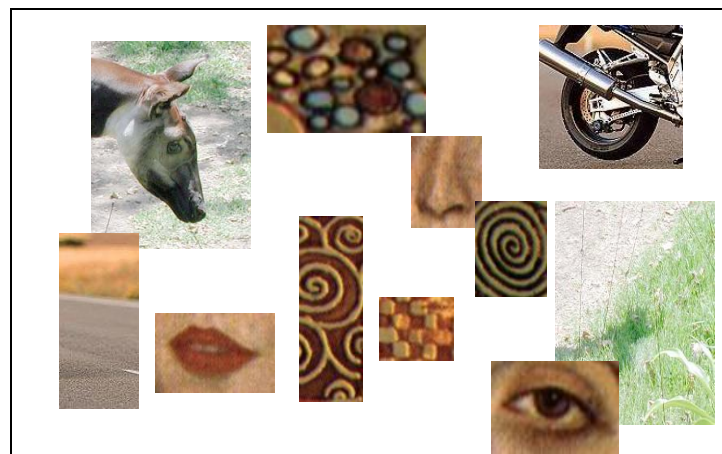
Zebra

Non-zebra





A collage of six images showing different parts of a zebra: a tail, a hindquarters, a side profile, a head, a front leg, and a close-up of the head.

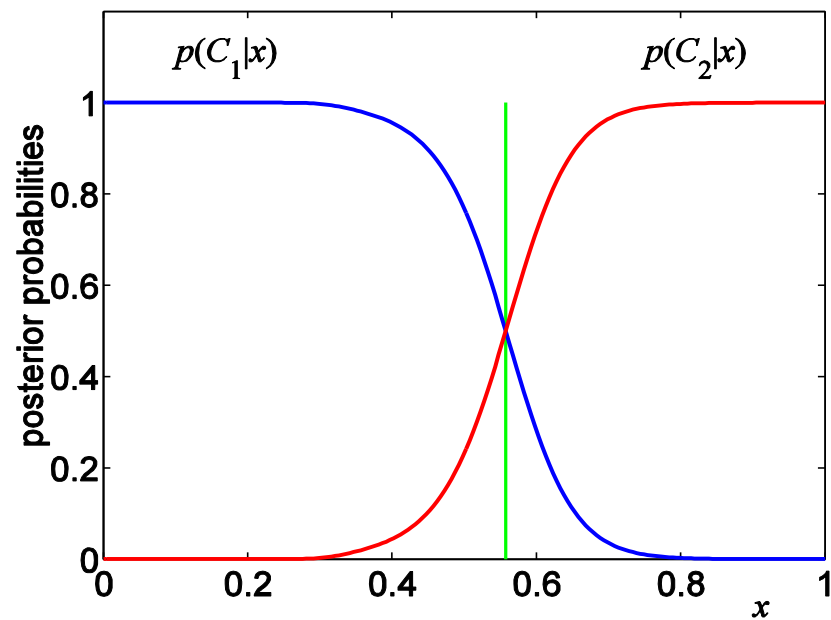
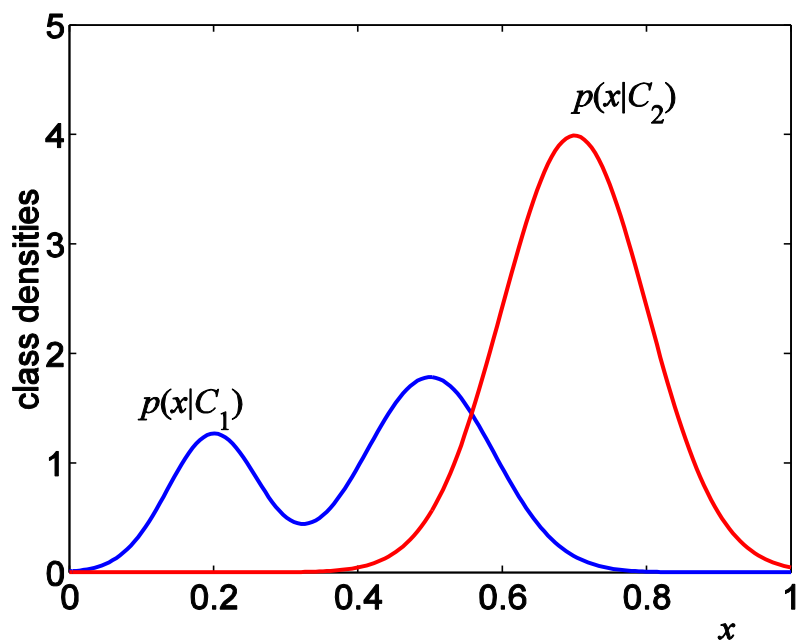


$p(image zebra)$	$p(image no\ zebra)$
Low	Middle
High	Middle $\rightarrow$ Low





目标检测识别中的几个问题——判别式与生成式

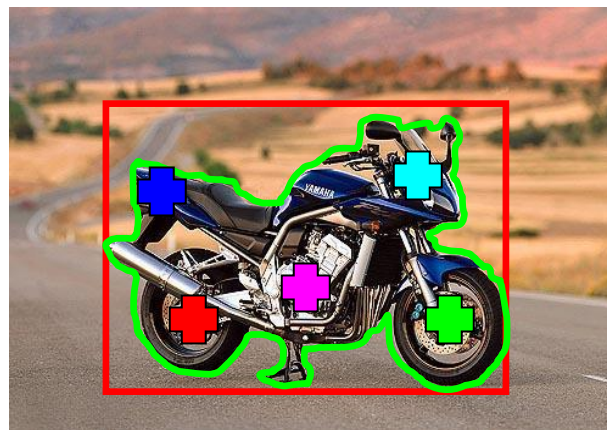




目标检测识别中的几个问题——其它

- 采用的监督
 - Manual segmentation; bounding box; image labels; noisy labels
- 批量学习/增量学习
- 训练图像:
 - 过拟合问题
 - 负样本问题

Contains a motorbike





作业

- 分析HOG、颜色直方图、PCA、shapecontext、特征点+BOW这几种表达的平移、旋转（平面内）、尺度、遮挡不变性，并分析原因

	平移	旋转	尺度	遮挡
HOG				
颜色直方图				
PCA				
shapecontext				
特征点+BOW				



The end !